Analysis of asphalt samples from the B30 cave

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1. Introduction:

On the 31/7/2023 we received 2 samples of asphalts that were scrapped from rock sickles. The samples are Q24 L IV B30-63 and Q24 L IV B30-65 (henceforth B30-63 and B30-65). The weight of the samples is 114.5 and 101mg respectively.

We were asked to compare the molecular composition of the B30 asphalts to an asphalt sample from the Nahal Heimar seep in attempt to establish chemical correlation between the samples.

2. Methods:

The studied asphalts were divided into four fractions according to their polarity: Saturates, Aromatics, Resins and Asphaltenes (known as "SARA"; Peters, et al., 2005). Asphaltenes precipitation was performed by mixing approximately 100 mg of sample with 50 times (by weight) *n*-pentane in a clean 20 ml vial. The vial was sealed, and its contents were ultrasonically mixed for 60 minutes. Following the mixing the vial was left overnight (at least 12 hours) to precipitate all asphaltenes. The remaining solution (maltene solution) comprising the saturate, aromatic and resin fractions was filtered and transferred for further separation by column liquid chromatography (LC). The column used was composed of pre-activated silica gel (2.5 g) and alumina (~0.2 g). The saturate fraction was eluted from the column with *n*-hexane (20mL), the aromatic fraction with dichloromethane (20 mL), and the resins with methanol (20 mL). The solvent used for each fraction (i.e., saturate, aromatic, resins) was evaporated under gentle flow of nitrogen and the fractional weight was recorded.

The saturate fraction of each asphalt was analysed using an Agilent 7890A GC coupled with an Agilent 5975C mass selective detector (GC-MS). A 30m·0.25mm·0.25µm DB-5MS fused silica column was used. About 1 µL of the saturate fraction was injected at split 10 and constant injector temperature of 320°C. Helium was used as a carrier gas at a constant rate of 1.5 ml/min. The temperature program used is 50°C for 4 minutes, a 10°C/min linear gradient up to 320°C with a hold at final temperature (320°C) for 20 minutes. This analysis provides accurate data regarding the molecular distribution of biomarkers in the steranes and hopanes groups.

3. Results and discussion:

All analyzed samples show a limited TIC envelope with significant unresolved complex-mixture (UCM) hump (Fig. 1), the Nahal Heimar asphalt seem to exhibit the most limited composition out of the three samples. The B30 asphalt samples contain traces of alkanes (Fig. 1), it appears that sample B30-63 may contain higher concentration of various alkanes. Due to their lability towards various degradation processes (water-washing, biodegradation, evaporation etc.), the presence of alkanes suggests sample B30-63 is less degraded than B30-65 which exhibits fewer alkanes. Both of the B30 samples are distinctively less degraded than the asphalt sample of Nahal Heimer (Fig. 1).

The hopanes envelopes of the B30 samples are characterized by C_{29}/C_{30} hopane ratio>1, high C_{35}/C_{34} homohopane ratio, very low Ts/Tm ratio, presence of tricyclic terpanes (mainly C_{23} tricyclic terpane) and presence of a peak tentatively identified as gammacerane which is lower than the adjacent C_{31} hopane isomer (Fig. 2). The Nahal Heimar asphalt sample bears much similarity to the B30 samples yet differs by having C_{29}/C_{30} hopane ratio<1, elevated C_{23} tricyclic terpane and higher ratio of gammacerane to its adjacent C_{31} hopane. The sterane fraction of all asphalt samples displays dominance of C_{28} steranes followed by C_{27} and C_{29} steranes respectively.

The main differences between the hopanes distribution in the studied samples and previously studied asphalts of the Dead Sea Basin are: A. The C_{29}/C_{30} hopane ratio is >1 in the B30 samples yet always ≤1 in the samples of the Dead Sea asphalts; B. The ratio of gammacerane to its adjacent C_{31} hopane is <1 in the B30 asphalts yet is >1 in Dead Sea asphalts; C. The presence of tricyclic terpanes in the B30 asphalts is mostly limited to the C_{23} tricyclic terpane which is less dominant than the TS peak. In contrast, all Dead Sea asphalts samples show a variety of tricyclic terpanes and have a very dominant C_{23} tricyclic terpane peak when compared to the TS peak (Rullkötter et al., 1985; Rullkötter and Nissenbaum, 1988; Connan et al., 1992; Connan and Nissenbaum, 2004; Connan et al., 2006). The steranes of the B30 asphalts shows simillar distribution to that identified in other asphalts and oils from the Dead Sea Basin (Rullkötter et al., 1985; Connan et al., 1992).

While showing distinct differences from the Dead Sea asphalts, the B30 asphalts bear great resemblance to oils from the Dead Sea Basin: virtually identical hopane distribution is known for the oils of the Dead Sea Basin (Spiro et al., 1983; Rullkötter et al., 1985) which are known to originate from the same source as the Dead Sea asphalts (Gardosh and Tannenbaum, 2014). The samples of crude oil that bear the most resemblance to the B30 samples were produced from depth of >1km and therefore were obviously inaccessible for the people who used the B30 asphalts. Overall, the similarities of the hopanes and steranes distribution between the B30 asphalts and the oil of the Dead Sea Basin hints they are genetically related and formed from a similar source rock.

A previous study that focused on trends of degradation in asphalts from the Dead Sea Basin suggested that the C_{29}/C_{30} hopane ratio is decreasing due to preferential degradation of the C_{29} hopane while the gammacerane/C_{31} hopane tends to increase during degradation as the C_{31} is prone to degradation (Connan et al., 2022). When compared to the Nahal Heimar asphalt, both B30 samples appear to be less degraded. Hence, it is unlikely that outcrop

samples of asphalt from Nahal Heimar were used for the sickles in B30 cave. On the other hand, previous study demonstrated variabilities in the composition of the Nahal Heimar asphalt over time (Rullkötter et al., 1985). Hence, it is possible that the B30 asphalts represent the composition of the Nahal Himer seep (or other seeps in the Dead Sea basin) at the time of sampling.

4. Conclusions:

The B30 asphalt samples display resemblance of biomarkers to both asphalts and oils from the Dead Sea basin, therefore indicating the B30 asphalts are genetically related to the different petroleum shows in the Dead Sea Basin and were collected locally. Nevertheless, the B30 asphalts exhibit some properties which suggest they were not collected from the Nahal Heimar seep which was hypothesized to be the source.

5. Figures:

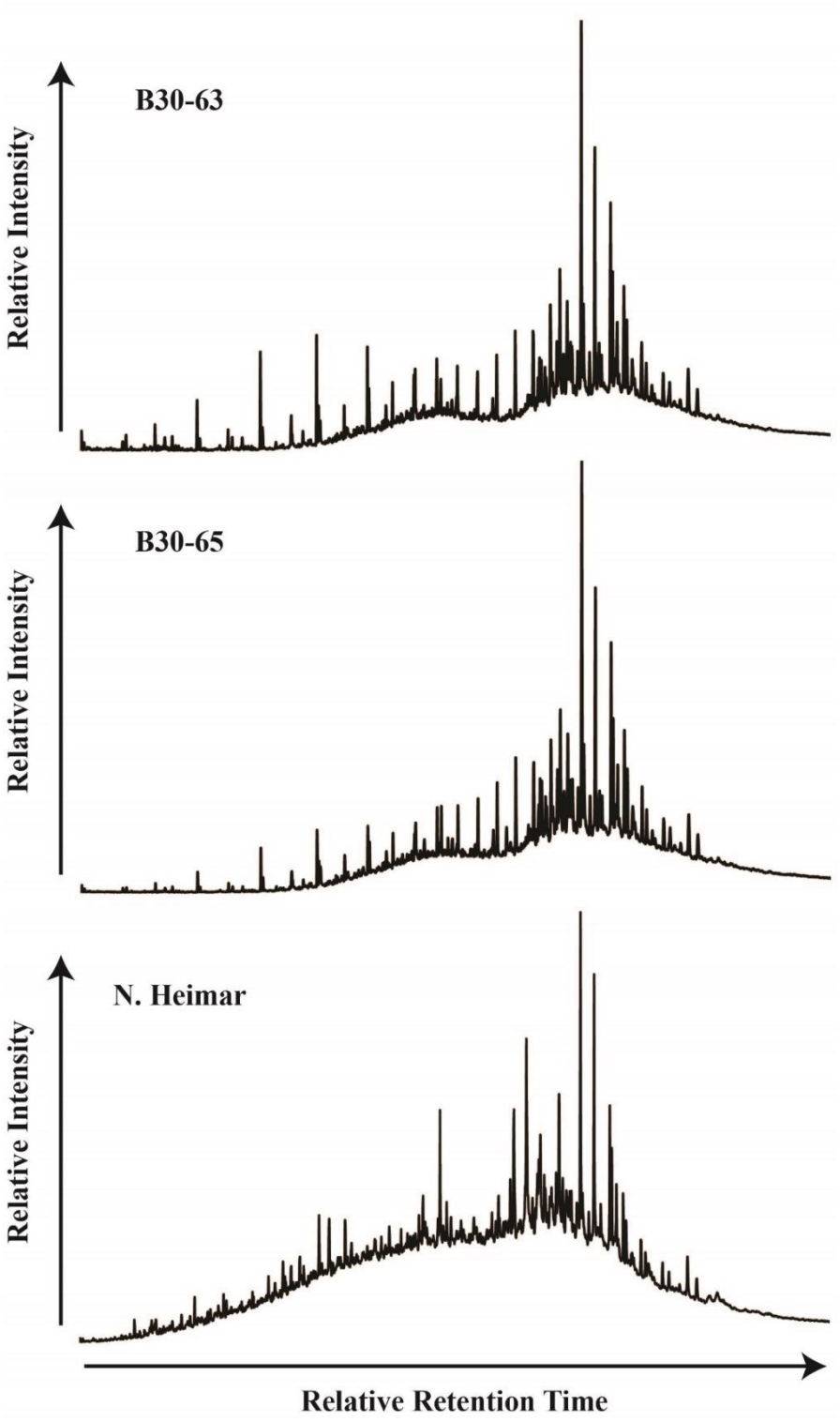

Figure 1- Total-ion current (TIC) GC-MS chromatograms of the saturate fraction of B30 and Nahal Heimar asphalts.

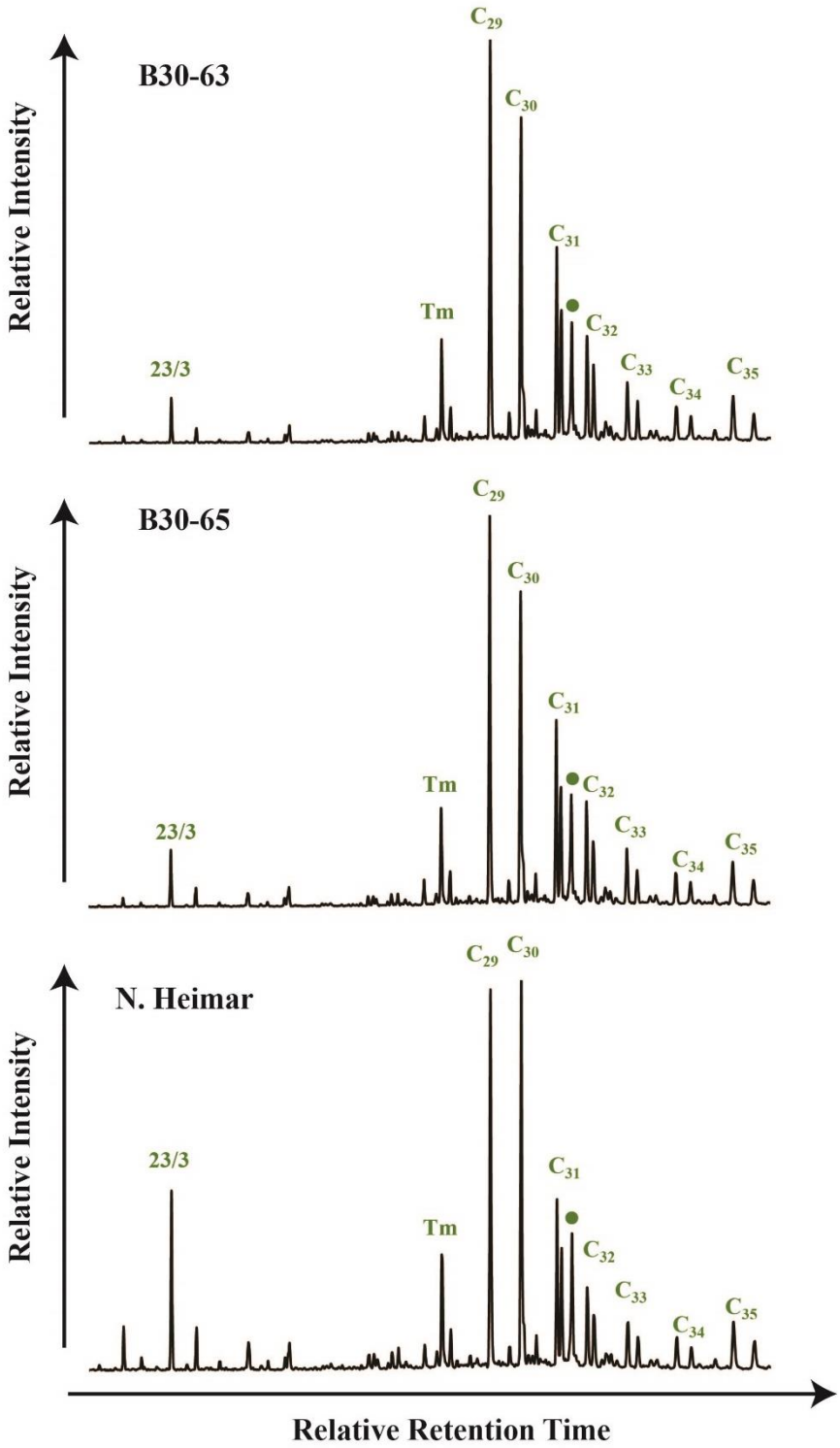

Figure 2- Partial GC-MS (SIM) chromatograms showing hopane (m/z 191) distribution in B30 and Nahal Heimar asphalts. Peak marked with ● is tentatively identified as Gammacerane.

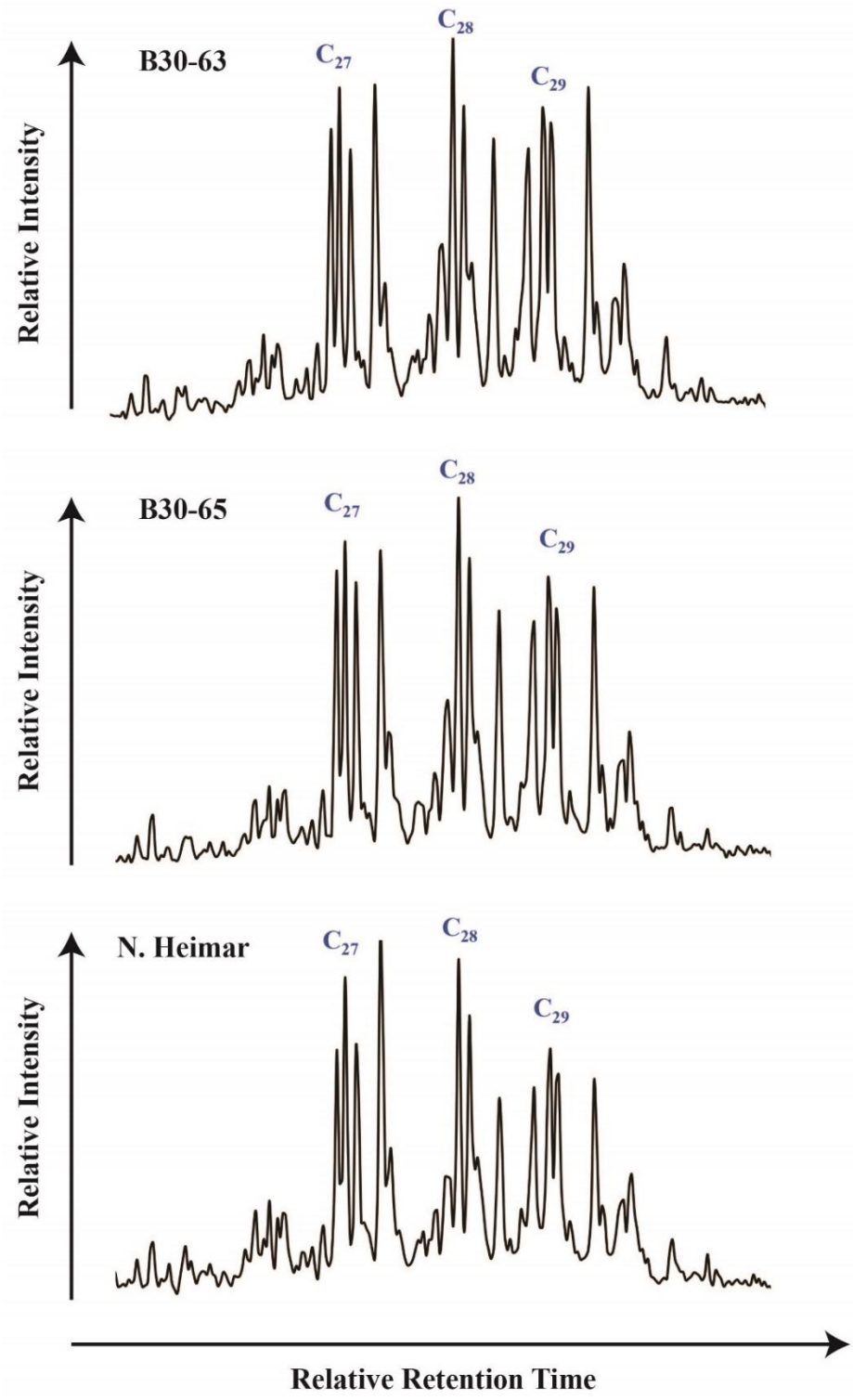

Figure 3- Partial GC-MS (SIM) chromatograms showing sterane (m/z 217) distribution in B30 and Nahal Heimar asphalts.

6. References:

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